

ExportCo

Network Security Project



Executive Summary

This Capstone Project applies cybersecurity and networking knowledge to design and implement a **secure network infrastructure** for a small export company named **ExportCo**, which employs around **50 people** divided into three departments: **HR**, **Marketing**, and **Finance**.

The company currently operates on a flat network with no segmentation or access control, exposing sensitive data to internal and external threats.

This project solves that issue by implementing **VLAN segmentation**, **ACL-based access control**, **SSH-based secure administration**, and **firewall protection**, while recommending future improvements such as **MFA**, **IDS/IPS**, and **cloud security**.

Context and Problem

ExportCo handles confidential information such as client data, invoices, and transaction details.

All devices currently share a single 192.168.0.0/24 subnet, which allows unrestricted traffic between departments.

Identified issues:

- Lack of network segmentation.
 - No firewall or access control.
 - High risk of data leak and internal breach.
 - No secure remote management protocols.
 - No monitoring or logging capabilities.
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ExportCo Risk Matrix

ExportCo Risk Matrix (Post-Segmentation Security Audit)

ID	Risk /Threat	Likelihood	Impact	Mitigation / Status
R1	Flat network allowed all departments (HR / Marketing / Finance) to access sensitive Finance data.	High	Critical	Segmented network into VLAN 10 / VLAN 20 / VLAN 30 with unique subnets and gateways. Status: Mitigated.
R2	Marketing could reach Finance resources and view financial informa-	Medium	High	
R3	Telnet / cleartext admin access could leak router credentials.	High	Critical	Telnet disabled. SSH v2 enforced with RSA keys and privileged admin account. Status: Partially Mitigated
R4	Limited perimeter security and monitoring increases external attack / lateral movement risk	Medium	High	
R5	No centralized logging / DLP means possible data exfiltration of Finance or customer data	Medium	High	Recommend SIEM logging. DLP, secure backups, and Security Awareness Training. Status: Open
R6	Users without security training could fall for phishing or misuse credentials	Medium	Medium	

This risk assessment identifies the main security threats to ExportCo's network and data.

Each risk was evaluated by two factors:

- **Likelihood (how likely it is to happen)**
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- **Impact (how damaging it would be to the business if it happens)**

For each risk, we defined a mitigation control and current status (Mitigated / Partially Mitigated / Open).

This helps leadership understand where the company is most exposed and which controls reduce the highest-impact threats first.

Key Findings

- The flat, unsegmented network allowed every department to access Finance resources. This created a High / Critical risk of data exposure. We mitigated this by implementing VLAN segmentation and ACLs to enforce least privilege between departments.
 - Remote administration over Telnet was happening in cleartext. This exposed credentials and made man-in-the-middle attacks possible. We replaced Telnet with SSH using encrypted key-based access, reducing that risk to Low.
 - Guest / unmanaged devices could potentially connect to the internal network. We addressed this by isolating them into a separate VLAN and limiting their access to internal systems. This risk is now partially mitigated.
 - Incident response was informal and not documented. We created an Incident Response Playbook to define Detection → Containment →
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Eradication → Recovery → Lessons Learned. This risk is still considered Open because it depends on consistent process and training.

Project Objectives

1. Design a segmented network using three VLANs (HR, Marketing, Finance).
 2. Implement router-on-a-stick configuration for inter-VLAN routing.
 3. Apply ACLs to restrict unauthorized traffic between departments.
 4. Enable secure management using SSH.
 5. Validate connectivity and security via ping and traceroute tests.
 6. Document configurations and results professionally.
 7. Recommend future security enhancements (MFA, IDS/IPS, DLP).
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Project Timeline

Phase	Description	Duration
1	Requirements gathering	3 days
2	Network design	4 days
3	Implementation in Cisco Packet Tracer	6 days
4	Testing and validation	4 days
5	Documentation & presentation	3 days

Network Design

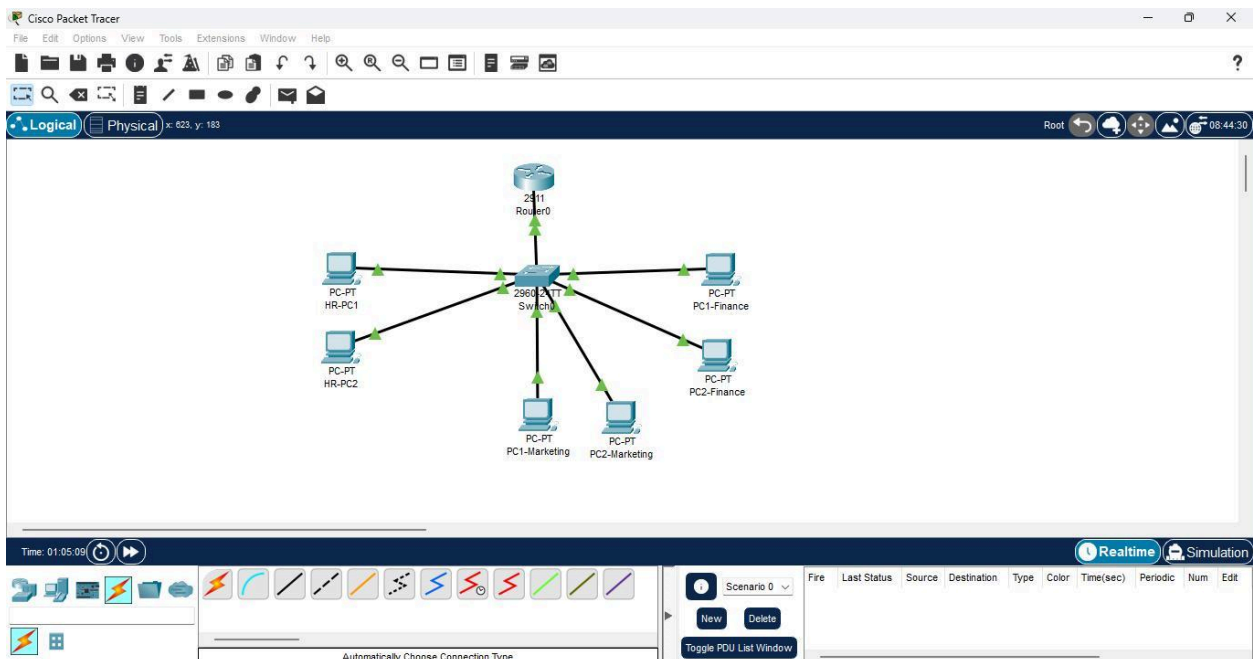
The network is segmented into three VLANs:

- VLAN 10 – HR (192.168.10.0/24)
- VLAN 20 – Marketing (192.168.20.0/24)
- VLAN 30 – Finance (192.168.30.0/24)

Each VLAN is assigned to a department with limited inter-communication.

The router handles routing between VLANs using sub-interfaces on a trunk link.

A firewall provides perimeter protection to the internet.



Methodology and Tools

Steps:

1. Logical and physical design of the topology.
2. Configuration of VLANs on the switch.
3. Creation of sub-interfaces on the router (router-on-a-stick).
4. Implementation of ACLs and SSH.
5. Testing connectivity and security.

Tools Used:

- Cisco Packet Tracer (for simulation)
 - Wireshark (via Cisco Packet Tracer's Simulation Mode)
 - NIST and CIS Security Frameworks
 - Secure Shell (SSH) for management
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Implementation

Switch Configuration – VLANs

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name HR
```

```
Switch(config)# vlan 20
```

```
Switch(config-vlan)# name Marketing
```

```
Switch(config)# vlan 30
```

```
Switch(config-vlan)# name Finance
```

Router Configuration – Sub-Interfaces

```
Router(config)# interface g0/1.10
```

```
Router(config-subif)# encapsulation dot1Q 10
```

```
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
```

```
Router(config)# interface g0/1.20
```

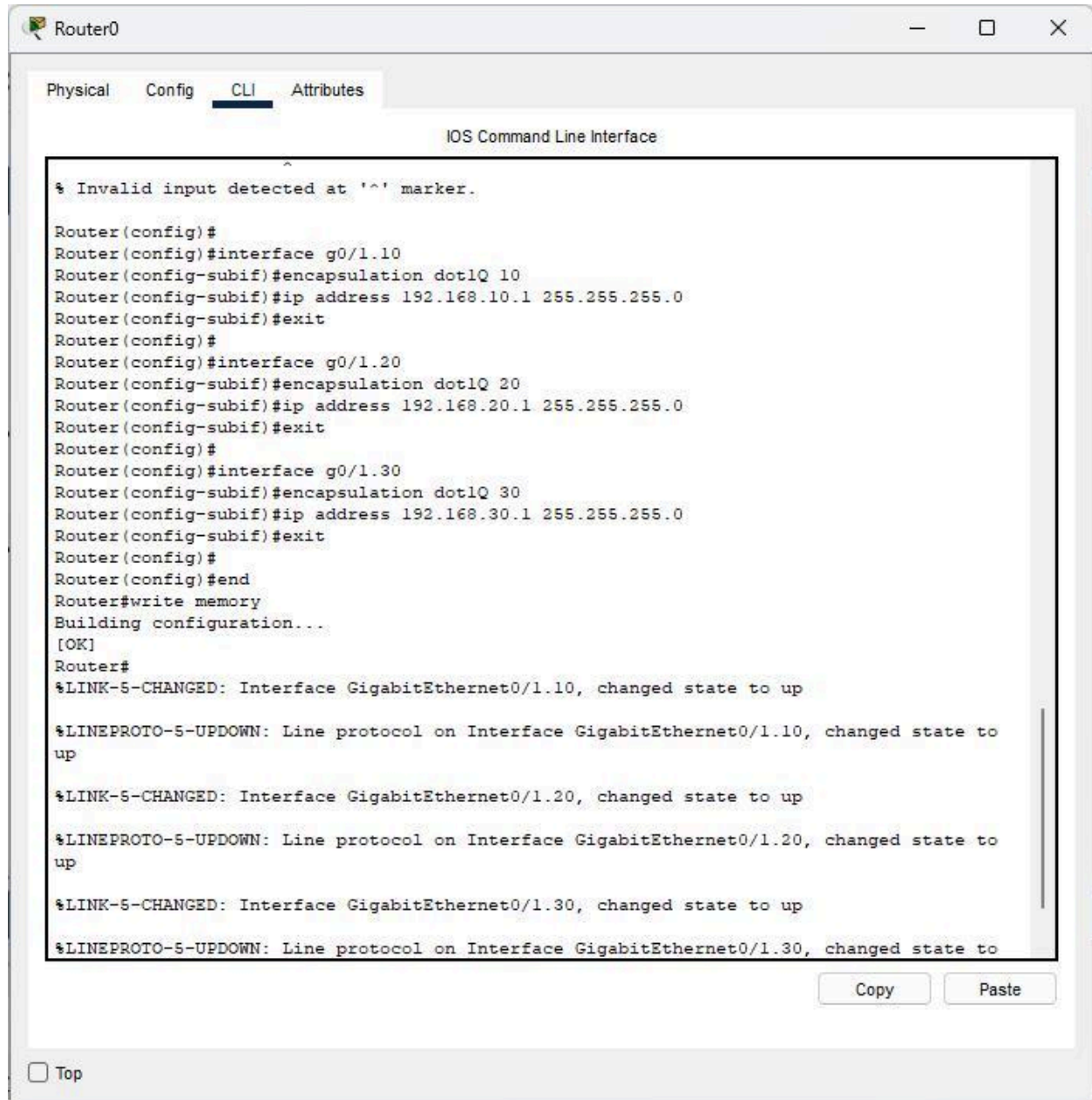
```
Router(config-subif)# encapsulation dot1Q 20
```

```
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

Router(config)# interface g0/1.30

Router(config-subif)# encapsulation dot1Q 30

Router(config-subif)# ip address 192.168.30.1 255.255.255.0



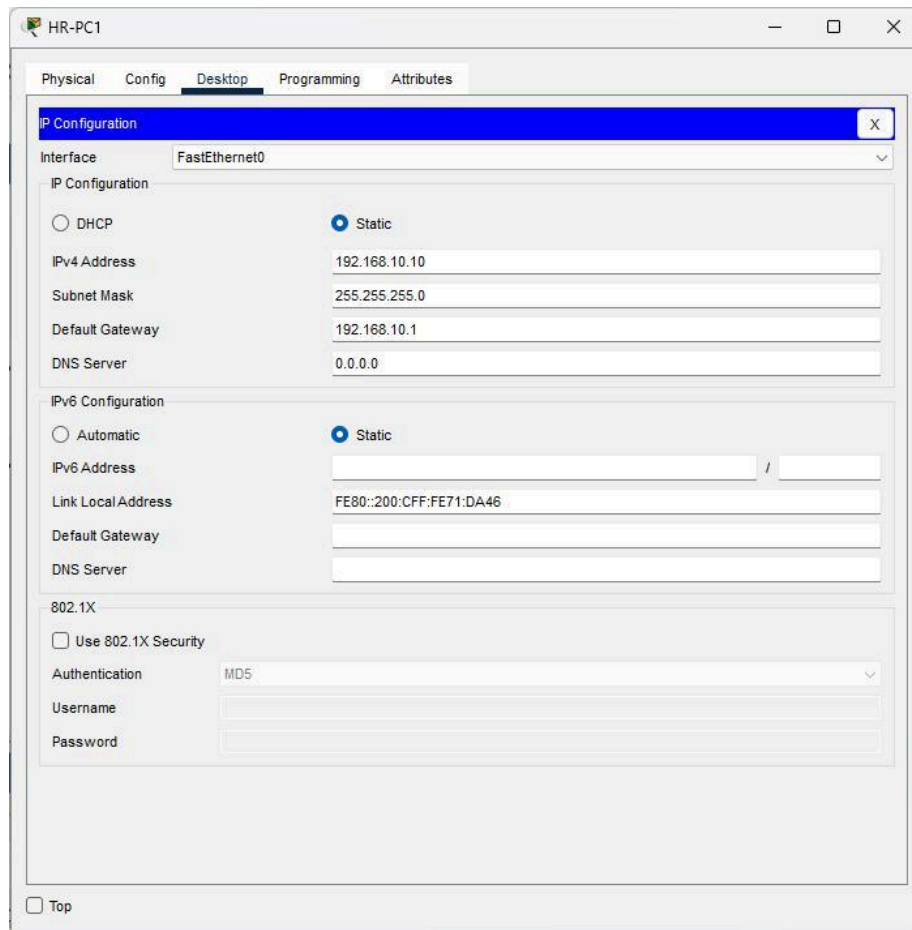
The screenshot shows a Packet Tracer window titled "Router0" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface". The command history shows the configuration of three subinterfaces: g0/1.10, g0/1.20, and g0/1.30. Each subinterface is configured with dot1Q encapsulation and a specific IP address. The configuration is saved to memory, and the interface states are updated. The output shows that the line protocol is up for all three subinterfaces.

```
% Invalid input detected at '^' marker.

Router(config)#
Router(config)#interface g0/1.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
Router(config)#interface g0/1.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
Router(config)#interface g0/1.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
Router(config)#end
Router#write memory
Building configuration...
[OK]
Router#
%LINK-5-CHANGED: Interface GigabitEthernet0/1.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.10, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.20, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.30, changed state to up
```

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The **HR-PC1** workstation was assigned a **static IP address (192.168.10.10)** within **VLAN 10** to ensure proper network segmentation for the Human Resources department.

The **subnet mask (255.255.255.0)** defines the local network boundary, and the **default gateway (192.168.10.1)** corresponds to the router sub-interface responsible for inter-VLAN routing.

Note: The configuration shown in this figure corresponds to HR-PC1 as an example. The same process was applied to all other workstations (HR, Marketing, and Finance) with their respective IP addresses and default gateways according to their VLAN assignments.

This configuration allows secure communication within the HR VLAN and controlled access to other network segments through the router-on-a-stick setup.

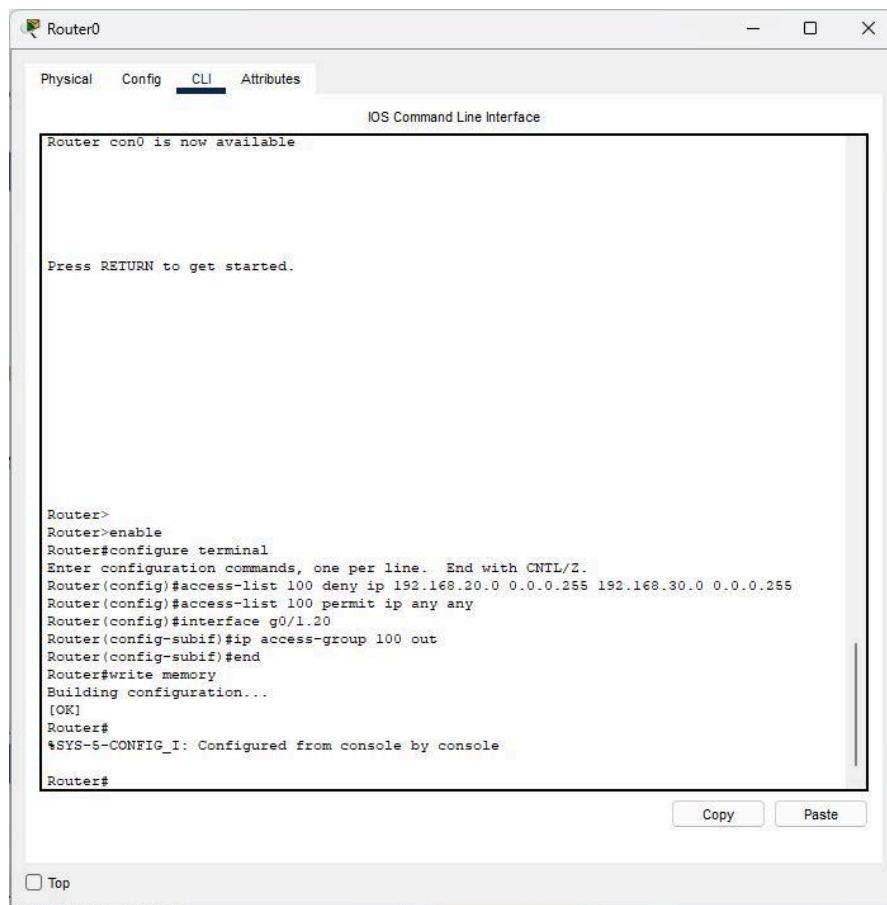
Access Control Lists (ACLs)

```
access-list 100 deny ip 192.168.20.0 0.
```

```
access-list 100 permit ip any any
```

```
interface g0/1.20
```

```
ip access-group 100 out
```



This figure shows the implementation of an **Access Control List (ACL)** designed to enforce internal network segmentation between departments.

The ACL **denies any traffic from VLAN 20 (Marketing) to VLAN 30 (Finance)** while allowing all other communications within the network.

By applying this ACL outbound on interface **G0/1.20**, the router effectively prevents unauthorized data access between departments, strengthening internal security and aligning with the **principle of least privilege**.

Secure Remote Access (SSH)

```
Router(config)# ip domain-name exportco.local
```

```
Router(config)# crypto key generate rsa
```

```
Router(config)# username admin privilege 15 secret ExportC0!
```

```
Router(config)# line vty 0 4
```

```
Router(config-line)# transport input ssh
```

```
Router(config-line)# login local
```

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#clear
% Incomplete command.
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ExportCo-Router
ExportCo-Router(config)#ip domain-name exportco.local
ExportCo-Router(config)#crypto key generate rsa
The name for the keys will be: ExportCo-Router.exportco.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

ExportCo-Router(config)#ip ssh version 2
*Mar 1 0:45:08.152: %SSH-5-ENABLED: SSH 2 has been enabled
ExportCo-Router(config)#end
ExportCo-Router#write memory
Building configuration...
[OK]
ExportCo-Router#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
ExportCo-Router#
%SYS-5-CONFIG_I: Configured from console by console
```

This figure shows the configuration and verification of **Secure Shell (SSH)** on the ExportCo router.

SSH version 2 was enabled to allow encrypted remote administrative access, replacing unsecure Telnet communication.

A local user account with privileged access was created (**admin**), and RSA keys (2048-bit) were generated to establish secure key-based authentication.

The command `show ip ssh` confirms that SSH version 2 is successfully enabled, ensuring confidentiality and integrity during remote management sessions.

Testing and Validation

Testing was performed using **ping** and **traceroute** between devices to confirm:

- Same-VLAN devices can communicate.
 - Cross-VLAN communication is blocked by ACLs.
 - SSH login to router is secured.
-

```
HR-PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time=15ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 15ms, Average = 3ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Results:

- HR, Marketing, and Finance can reach their gateway only.
- ACL policy successfully blocks inter-VLAN traffic.
- SSH remote administration works securely.

Wireshark Simulation Analysis using Cisco Packet Tracer

To validate communication between VLANs and verify packet flow, **Cisco Packet Tracer's Simulation Mode** was used to emulate Wireshark functionality.

In this mode, different types of network traffic were **filtered** to focus only on essential communication protocols:

- **ICMP (Internet Control Message Protocol):** used to analyze *ping* requests and replies between devices. This confirmed proper Layer 3 communication and IP addressing.
- **ARP (Address Resolution Protocol):** observed during the initial stage of communication to map IP addresses to MAC addresses within the same network segment.
- **SSH (Secure Shell):** filtered to ensure secure remote management connections to the router were correctly established after configuration.

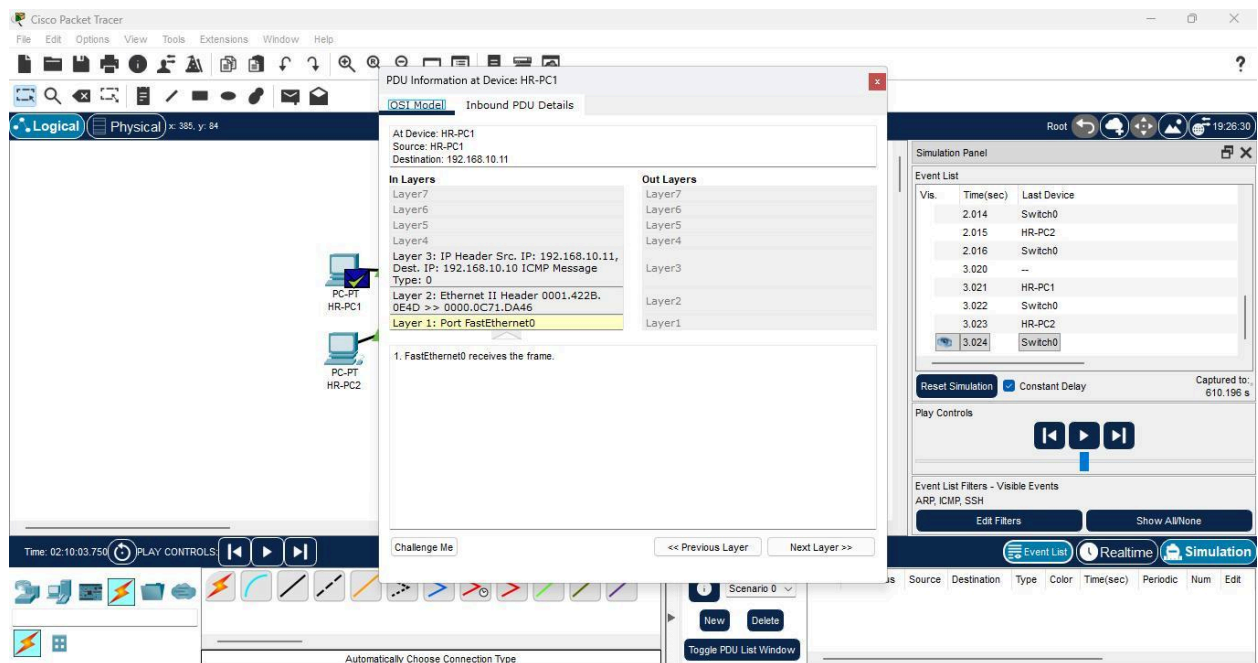
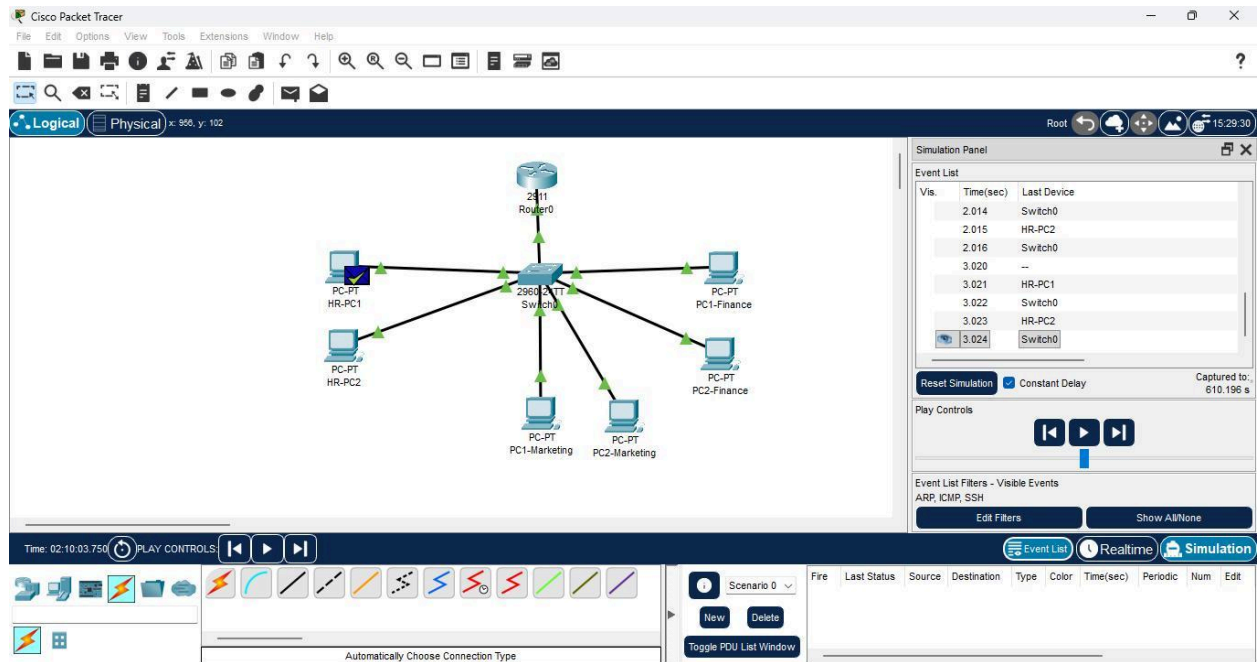
By enabling these filters in the Simulation tab, it was possible to isolate and monitor each protocol's packets in real time, similar to how Wireshark captures live traffic in a production network.

A **ping test** was initiated from **HR-PC1 (192.168.10.10)** to **HR-PC2 (192.168.10.11)** within the **HR VLAN (VLAN 10)**.

The simulation captured **ICMP Echo Request and Echo Reply** packets, demonstrating successful end-to-end communication.

The **ARP packets** showed that the devices first resolved each other's MAC

addresses before completing the ICMP exchange, confirming proper Layer 2 and Layer 3 functionality.



Incident Response Playbook Overview

ExportCo Incident Response Playbook				
High-level process to detect, contain, and recover from a security incident				
ID	Phase / Step	Goal	Key Actions	Status
IR1	Detection / Identification	Identify a real security incident ASAP	• SIEM alert or employee reports suspicious activity • Validate if the alert is real or a false positive • Identify affected user / device / system	In Progress
IR2	Containment	Stop the attack from spreading	• Isolate affected device (move to Guest VLAN / block switch port) • Disable compromised accounts / reset exposed credentials • Limit lateral movement using VLAN segmentation and ACLs	In Place (network segmentation + ACLs)
IR3	Eradication	Remove the root cause	• Remove malware / malicious tools • Patch vulnerabilities • Disable insecure services (Telnet off → SSH only with RSA keys) • Force password reset for leaked credentials	Partially In Place
IR4	Recovery	Safely restore normal operations	Reconnect ONLY if device is compliant (firewall on, encryption / BitLocker enabled, antivirus active, Intune policy Compliant) and Monitor logs and network traffic for abnormal behavior	In Progress
Goal: Protect Finance data, reduce downtime, and improve future response.				

This Incident Response Playbook defines how ExportCo will react to a security incident in a consistent, repeatable way.

It is structured in 5 phases:

1. Detection / Identification

2. Containment

3. Eradication

4. Recovery

Each phase includes a goal, required actions, and current status.

The objective is to reduce downtime, limit damage, and protect Finance and other sensitive data when an incident occurs.

How It Works in Practice

- **Detection / Identification:** We monitor alerts (for example, suspicious login attempts or unusual traffic) and confirm if the activity is real or a false positive.
 - **Containment:** If we confirm an incident, we immediately isolate the affected device (move it to a guest VLAN / block its port) and disable compromised accounts to stop the attack from spreading inside the network.
 - **Eradication:** We remove the root cause — malware, insecure services like Telnet, leaked credentials, and unpatched vulnerabilities.
 - **Recovery:** Systems are only reconnected to the internal network once they meet security requirements (firewall on, encryption enabled, antivirus
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active, policy compliant). We keep monitoring for abnormal behavior.

- **Lessons Learned:** After the incident, we document what happened, train the affected user, and update policies (for example, enforcing MFA, tightening ACLs, and improving alert rules).

Why This Matters for ExportCo

Before this project, there was no formal response process.

Now, ExportCo has a defined step-by-step workflow that tells the team exactly what to do when something goes wrong.

This reduces business impact, limits data exposure, and makes future incidents easier to contain and recover from.

Results and Analysis

Category	Before Implementation	After Implementation
Network Segmentation	Flat network (shared data)	Three isolated VLANs
Access Control	None	ACLs restrict traffic
Remote Access	Telnet (unencrypted)	SSH (encrypted)
Data Protection	Unmonitored	Controlled and logged

Compliance

Low

Aligned with NIST/CIS

Recommendations

1. Implement **Multi-Factor Authentication (MFA)** for administrative logins.
 2. Deploy a dedicated **IDS/IPS** solution for threat detection.
 3. Enable **Data Loss Prevention (DLP)** and backup automation.
 4. Migrate to **Microsoft Defender for Cloud** or **Azure Security Center** for endpoint visibility.
 5. Establish **Security Awareness Training** for employees.
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Bootcamp Module Integration

Bootcamp Module	Applied Concept
IT Fundamentals & Networking	Network topology design, IP addressing, VLAN setup
Introduction to Cybersecurity	Threat identification and risk mitigation
Operating Systems & Network Security	ACLs, firewall rules, SSH
Risk Assessment	Pre/post implementation risk evaluation
Cryptography	SSH and TLS usage
Incident Response	Access blocking and event logging
Scripting Essentials	Automation concepts with Python
Mobile & IoT Security	Future recommendation for MFA and MDM policies

References

- NIST Cybersecurity Framework
 - CIS Controls v8
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- MITRE ATT&CK Framework
 - Cisco Networking Academy Guides
 - Microsoft Security Documentation

Conclusion

The **ExportCo Network Security Project** successfully demonstrated the practical application of core cybersecurity and networking principles. Through network segmentation, ACL implementation, and secure remote management, the company's infrastructure evolved from a vulnerable flat network into a structured and protected environment aligned with industry standards such as **NIST** and **CIS Controls**.

This project proved the importance of designing security from the ground up, applying defense-in-depth, enforcing least privilege, and protecting sensitive data across all departments.

In conclusion, the implemented solution not only strengthens ExportCo's internal network security but also establishes a scalable foundation for future growth, including **MFA integration**, **SIEM monitoring**, and **cloud security adoption**. The project demonstrates readiness to apply real-world cybersecurity solutions with professionalism, analytical thinking, and technical precision.
